

Dynamic Multi-Sensor GeoAI Framework for Monsoon-Driven Landslide Susceptibility Evolution and Cross-Basin Transferability in Nepal

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Abstract

Landslide susceptibility models in Nepal are predominantly static—trained once on historical inventories and applied without updating as monsoon dynamics evolve. This study proposes a dynamic, multi-sensor GeoAI framework that integrates Sentinel-1 SAR pre/post-monsoon backscatter change, Sentinel-2 monsoon composites, NASADEM-derived topographic indices (TWI, SPI, curvature), and GPM IMERG rainfall extremes to map time-sensitive landslide susceptibility across the Arun Basin (2021–2025). Quantile Mapping (QM) bias correction aligns IMERG estimates with DHM Nepal gauge records to preserve extreme-event rainfall signatures. An XGBoost ensemble model is trained with SHAP-based explainability to identify dominant conditioning and triggering factors per terrain class. A Few-Shot Transfer Learning experiment then tests whether the Arun-trained model, fine-tuned on only 5% of labeled data, achieves competitive accuracy in the morphologically distinct Trishuli Basin—directly addressing the critical challenge of inventory-scarce basins. This framework directly supports Nepal’s National Landslide Risk Management Strategy and demonstrates open-data reproducibility for the broader Hindu Kush Himalayan region.

1 Introduction and Rationale

Nepal’s mountainous terrain hosts some of the world’s highest landslide densities, with the monsoon season (June–September) triggering over 80% of annual landslide fatalities. The Arun and Trishuli river basins represent contrasting geomorphological regimes—Arun’s deeply incised gorges with thin colluvial soils versus Trishuli’s more structurally complex mid-hill geology—making them ideal paired study areas for transferability analysis.

Despite substantial advances in Machine Learning-based Landslide Susceptibility Mapping (LSM), three critical gaps persist:

1. **Temporal Rigidity:** Most models treat susceptibility as static, ignoring the progressive soil saturation and LULC change that unfolds across seasons (2021–2025 captures the catastrophic September 2024 event).
2. **SAR Underutilization:** Sentinel-1 backscatter has rarely been used as a moisture proxy in Nepal. Raw imagery without Radiometric Terrain Flattening (RTF) produces false signals in high-relief terrain.
3. **Generalization:** Models trained on well-inventoried basins rarely transfer to data-scarce areas. Few-Shot domain adaptation offers a practical path for operational deployment.

2 Literature Review and Research Gaps

Recent assessments of the September 2024 rainfall events in Central Nepal documented over 3,200 landslides in six weeks, highlighting the need for near-real-time susceptibility updates (Lamichhane *et al.*, 2025). While ensemble ML achieves high accuracy for co-seismic events (Pyakurel *et al.*, 2024), these remain event-specific snapshots. Integrating dynamic factors like accumulated rainfall and LULC change significantly improves temporal accuracy (Lee *et al.*, 2022; Ye *et al.*, 2025), yet the role of Sentinel-1 SAR backscatter change as a moisture proxy remains an open frontier (Nava *et al.*, 2025).

For high-relief terrain, RTF correction (Mullissa *et al.*, 2021) is essential to eliminate topographic artifacts. Furthermore, Explainable AI via SHAP (Hussain *et al.*, 2025) has emerged as the standard for validating GeoAI outputs against geomorphological domain knowledge. Finally, addressing transferability using domain adaptation (Su *et al.*, 2024; Yu *et al.*, 2025) is critical for "no sample" basins in the Himalayas.

3 Methodology

3.1 Data Fusion and Feature Engineering

We utilize an entirely open-access stack including NASADEM (30m), Sentinel-2 (10m), Sentinel-1 GRD (10m), and GPM IMERG (0.1°).

- **Dynamic SAR Feature:** Pre-to-peak monsoon VV backscatter delta (ΔVV) is computed after RTF correction to capture soil moisture evolution.
- **Rainfall Fusion:** Quantile Mapping (QM) aligns IMERG estimates with DHM Nepal records to correct the systematic underestimation of extreme events in the Himalayas (Nepal *et al.*, 2021; Nair *et al.*, 2025).

3.2 Modeling and Transfer Learning

An **XGBoost Ensemble** optimized via Bayesian Optimization (Optuna) is trained on the Arun Basin. We implement SHAP to identify feature dominance. Transferability is tested via **Few-Shot Fine-Tuning**, using 5% of Trishuli labels to adapt the Arun-trained model via Feature-Aligned Transfer.

4 Timeline and Expected Impact

Month 1: GEE extraction, RTF correction, and QM bias correction.

Month 2: Model training and SHAP explainability analysis.

Month 3: Transfer learning experiments and validation on 2024 extreme events.

Month 4: Abstract submission (20 Nov 2026) and final manuscript preparation.

The primary outcome is a scalable, reproducible GeoAI pipeline for the Hindu Kush Himalaya, providing physically consistent evidence for operational early warning system design under Nepal's NDRRMA.

5 Bibliography

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